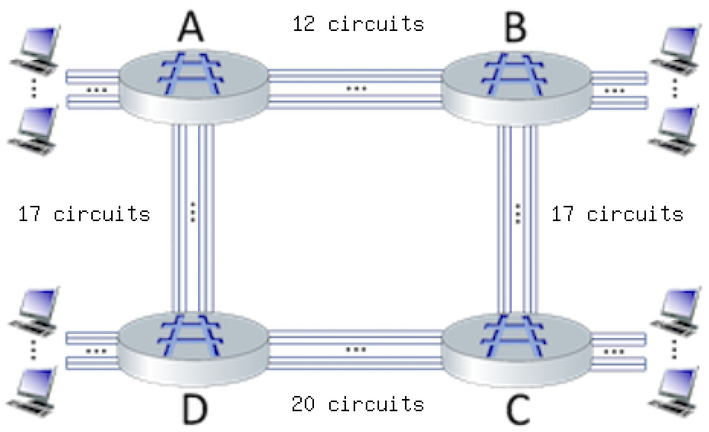
INTERACTIVE END-OF-CHAPTER EXERCISES

CIRCUIT SWITCHING

Consider the circuit-switched network shown in the figure below, with circuit switches A, B, C, and D. Suppose there are 12 circuits between A and B, 17 circuits between B and C, 20 circuits between C and D, and 17 circuits between D and A.



QUESTION LIST

1. What is the maximum number of connections that can be ongoing in the network at any one time?

2. Suppose that these maximum number of connections are all ongoing. What happens when another call connection request arrives to the network, will it be accepted? Answer Yes or No

3. Suppose that every connection requires 2 consecutive hops, and calls are connected clockwise. For example, a connection can go from A to C, from B to D, from C to A, and from D to B. With these constraints, what is the is the maximum number of connections that can be ongoing in the network at any one time?

4. Suppose that 20 connections are needed from A to C, and 11 connections are needed from B to D. Can we route these calls through the four links to accommodate all 31 connections? Answer Yes or No

SOLUTION

1. The maximum number of connections that can be ongoing at any one time is the sum of all circuits, which happens when 12 connections go from A to B, 17 connections go from B to C, 20 connections go from C to D, and 17 connections go from D to A. This sum is 66.

2. No, it will be blocked because there are no free circuits.

3. There can be a maximum of 32 connections. Consider routes A->C and C->A, sum the bottleneck links, consider any leftover capacity that would allow for B->D and D->B connections, and compare that value to the equivalent of B->D and D->B.

4. Using our answer from question 4, the sum of our needed connections is 31, and we have 32 available connections, so it is possible.

QUESTION 1 OF 4

What is the maximum number of connections that can be ongoing in the network at any one time?

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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